

Software Requirements Specification (SRS)

Project: Customer Chat Support System (CCSS)

1. Introduction

1.1 Purpose

This document details the functional and non-functional requirements for the **Customer Chat Support System (CCSS)**. This system enables real-time messaging between web application customers and support agents. [[1](#), [2](#), [3](#), [4](#)]

1.2 Scope

The CCSS is a web-based communication platform. It features a customer-facing chat widget and an internal agent dashboard. It manages live messaging, basic chat routing, and message log preservation. It does not cover video calls or third-party CRM integrations. [[1](#), [2](#), [3](#), [4](#), [5](#)]

2. Overall Description

2.1 Product Perspective

The system operates as an independent subsystem embedded within the company's existing website infrastructure. [[1](#), [2](#)]

2.2 User Classes

- **Customers:** Website visitors seeking instant assistance.
 - **Support Agents:** Internal staff responding to incoming customer issues.
 - **Administrators:** Managers tracking system metrics and configuring basic settings. [[1](#), [2](#), [3](#), [4](#), [5](#)]
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3. Functional Requirements

3.1 Customer Interface

- **Widget Activation:** The system must load a floating chat icon on the bottom right corner of the website.
- **Pre-Chat Form:** The system must prompt anonymous users for their name and email address before starting a session.
- **Message Exchange:** The system must transmit text messages instantly between the customer and the assigned agent.
- **Status Indicator:** The widget must display when an agent is typing or goes offline. [[1](#), [2](#), [3](#), [4](#), [5](#)]

3.2 Agent Dashboard [[1](#)]

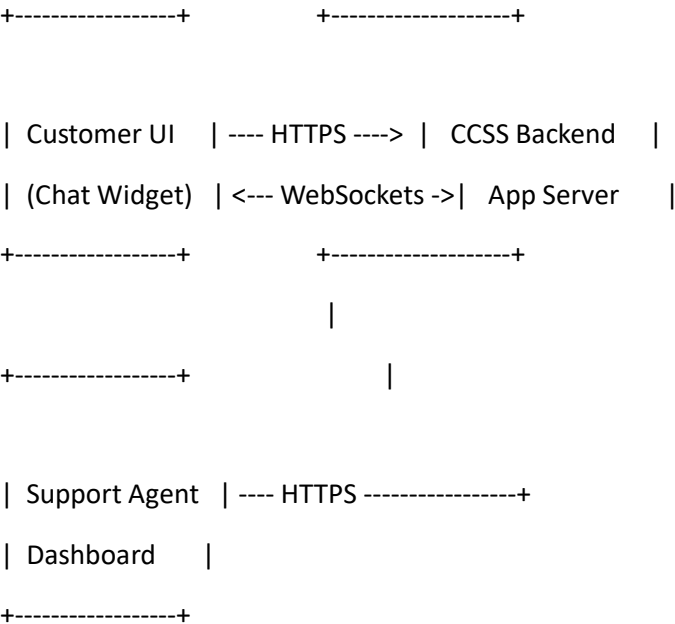
- **Queue Management:** The system must display all incoming, unassigned chat requests in a central queue.

- **Chat Acceptance:** The system must assign a chat session to an agent immediately when they click "Accept".
- **Canned Responses:** The dashboard must allow agents to select and send pre-written templates.
- **Session Termination:** The system must archive the session and clear the workspace when the agent clicks "End Chat". [[1](#), [2](#), [3](#), [4](#), [5](#)]

3.3 Administrative Features

- **User Management:** Administrators must have permissions to add, edit, or deactivate agent credentials.
- **Chat History Access:** The system must store searchable text logs of all completed chat sessions. [[1](#), [2](#)]

4. External Interface Requirements



4.1 User Interfaces

- **Widget Design:** Clean, minimal chat box responsive on both desktop browser viewports and mobile layouts.
- **Dashboard Layout:** Two-column console displaying the active chat queue on the left and the active message interface on the right. [[1](#)]

4.2 Software Interfaces

- **Network Protocol:** Utilizes WebSockets for duplex real-time messaging, falling back to HTTPS polling if blocked.
- **Database:** Connects to a relational database to store message histories, agent status, and user profiles. [[1](#), [2](#), [3](#), [4](#)]

5. Non-Functional Requirements

5.1 Performance

- **Latency:** Messages must display on the receiver's screen within 2 seconds of transmission.
- **Capacity:** The server backend must support up to 150 concurrent active chat sessions without performance degradation. [[1](#), [2](#)]

5.2 Security

- **Encryption:** All data in transit must run over HTTPS and secure WebSockets (WSS).
- **Authentication:** Support agents must log in using password hashing (BCrypt) before accessing customer data. [[1](#)]

5.3 Availability

- **Uptime:** The system must achieve a 99.5% operational uptime during standard corporate service hours.